

Foundations of Arithmetic Power Geometry: Local Parameter Deformations and Entropy at the Euclidean Target

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Abstract

We present Arithmetic Power Geometry (APG). This axiomatic local deformation framework reframes the algebraic solvability of power-type Diophantine equations as a problem of localized functional deformation. Global infimum calculations over infinite discrete subsets are often susceptible to asymptotic collapse arising from approximation phenomena. To mitigate this, APG isolates algebraic deformation locally around a fixed coordinate baseline designated as the Euclidean Target. We formalize the set-theoretic structure of a localized parameter space, the Power-Deformation Space. This space treats the exponent as a continuous deformation parameter. We state and prove the First-Order Closure-Obstruction Asymptotic Expansion Theorem. This theorem demonstrates that the first-order growth of the normalized closure defect away from the classical baseline ($n = 2$) possesses a leading-order coefficient proportional to the Shannon entropy of the normalized base weights. Finally, we establish the formal scope and analytical boundaries of the framework to ensure mathematical and conceptual clarity.

Keywords: arithmetic geometry; Diophantine approximation; exponent deformation; information theory; entropy; asymptotic analysis.

1 Introduction and Core Philosophy

In classical mathematics, algebraic equations are typically investigated through the structural and geometric properties of their solution sets. This methodology is formalized within algebraic geometry (Hartshorne, 1977) and arithmetic geometry (Hindry & Silverman, 2000; Silverman, 2007). These disciplines study geometric objects defined by polynomial equations and the arithmetic properties of rational and integral points lying on such objects (Lang, 2002; Bombieri & Gubler, 2006). Rather than focusing exclusively on the geometry of solution spaces, modern arithmetic investigations often seek structural invariants that characterize the behavior of algebraic relations across different mathematical domains.

We propose Arithmetic Power Geometry (APG) as an axiomatic local deformation framework for investigating a distinct structural phenomenon: algebraic closure under continuous exponent variation. The central question of APG is not what geometric object a particular equation defines, but how closure relations evolve when the exponent is treated as a continuously varying parameter rather than a fixed constant. In this setting, exponentiation functions as a deformation mechanism acting upon fixed positive base coordinates.

The motivation for this perspective arises from the contrast between classical Euclidean closure and higher-power relations. The equation $a^2 + b^2 = c^2$ admits infinitely many integral solutions, whereas power-type equations with larger exponents exhibit substantially different arithmetic behavior. APG does not attempt to replace existing theories in arithmetic geometry, Diophantine geometry, or algebraic geometry. Instead, it provides a localized analytical framework for studying how closure behavior changes as the exponent moves away from a distinguished baseline.

By shifting attention from varying coordinates under fixed exponents to varying exponents over fixed coordinates, APG connects naturally with mathematical disciplines that investigate structural change through parameter variation. Examples include information geometry (Amari, 2016), deformation theory, and tropical geometry (Maclagan & Sturmfels, 2015). Within this framework, closure is viewed not merely as a binary property of existence or non-existence, but as a quantity whose local sensitivity to exponent variation can be analyzed mathematically.

The present paper develops this framework in a localized setting centered on the Euclidean baseline ($n = 2$). A normalized closure defect functional is introduced to quantify deviations from perfect closure, and a first-order asymptotic theorem is established linking the resulting local deformation behavior to the Shannon entropy of normalized coordinate weights. The goal is not to derive global number-theoretic consequences, but to provide a rigorous analytical language for studying exponent deformation and localized closure phenomena.

2 Foundational Axioms of APG

To establish a formal analytic environment for this framework, we define a localized parameter space in which exponentiation acts as a continuous deformation operator.

Axiom 1. The Power-Deformation Space, $\mathcal{P}(a, b)$. For any fixed positive real base coordinates $(a, b) \in (\mathbb{R}^+)^2$, there exists a localized two-dimensional parameter space called the Power-Deformation Space, denoted by $\mathcal{P}(a, b)$, and defined as

$$\mathcal{P}(a, b) = \{(c, n) \in \mathbb{R}^+ \times \mathbb{R}^+ : c > 0, n > 0\}. \quad (1)$$

A point $(c, n) \in \mathcal{P}(a, b)$ represents a target scalar coordinate c evaluated under a continuous exponent deformation parameter n .

Axiom 2. Power Composition and Closure. Elements within the parameter space $\mathcal{P}(a, b)$ are governed by an implicit composition rule. Perfect algebraic closure is achieved at a given parameter value n if and only if there exists a coordinate c within the background universe such that

$$a^n + b^n = c^n \quad (2)$$

holds exactly.

Axiom 3. The Multi-Universe Domain. The manifestation of closure depends on the chosen background algebraic universe U , where

$$U \in \{\mathbb{Z}^+, \mathbb{Q}^+, \mathbb{R}^+, \mathbb{C}\}, \quad (3)$$

following standard conventions in arithmetic geometry (Hindry & Silverman, 2000; Silverman, 2007).

We define the binary valuation map

$$\text{Closure}_U(E) = \begin{cases} 1, & \text{if } E \text{ closes perfectly inside } U, \\ 0, & \text{if } E \text{ fails to close inside } U. \end{cases} \quad (4)$$

This mapping records whether exact closure exists within the selected algebraic universe.

Axiom 4. The Normalized Closure Defect, d_P . We endow the Power-Deformation Space $\mathcal{P}(a, b)$ with a localized closure-defect functional that measures deviation from perfect closure. For a given point $(c, n) \in \mathcal{P}(a, b)$, define

$$d_P(c, n) = \frac{|c^n - a^n - b^n|}{c^n}, \quad (5)$$

where $d_P = 0$ indicates an unobstructed and perfectly closed configuration.

The quantity d_P is a normalized defect functional rather than a metric in the differential-geometric sense. It measures relative departure from exact closure and shares structural similarities with error functionals commonly encountered in Diophantine approximation (Bugeaud, 2004; Schmidt, 1980).

Axiom 5. The Power Deformation Parameter, K_P . We define

$$K_P(n) = n - 2 \quad (6)$$

and refer to K_P as the Power Deformation Parameter.

The exponent $n = 2$ serves as the distinguished Euclidean baseline of the framework.

- When $K_P = 0$ (equivalently $n = 2$), the system coincides with the classical Euclidean closure configuration.
- When $K_P \neq 0$, the exponent departs from the Euclidean baseline and acts as a deformation parameter, introducing localized structural strain into the closure relation.

The choice $K_P = n - 2$ provides a natural local coordinate centered at the Euclidean exponent and serves as the primary deformation variable throughout the remainder of the framework.

3 Local Defect Fields and the Baseline Target

To preserve analytical clarity and avoid the asymptotic collapse that can arise in global infimum constructions over infinite coordinate domains, APG studies deformation behavior locally

around a fixed reference coordinate. This localization is motivated by classical results in Diophantine approximation, where sufficiently large integers may approximate continuous structures with arbitrarily high precision without producing exact closure (Schmidt, 1980; Bugeaud, 2004). Consequently, the framework focuses on a distinguished baseline coordinate rather than attempting to construct a global obstruction invariant.

The Euclidean Target. For any point in the Power-Deformation Space $\mathcal{P}(a, b)$, the distinguished baseline coordinate is chosen to be the classical Pythagorean closure point

$$c_2 = \sqrt{a^2 + b^2}. \quad (7)$$

The coordinate c_2 is referred to as the Euclidean Target.

The Euclidean Target serves as the historical and analytical reference point of the framework. It represents the unique coordinate at which perfect closure occurs for the classical exponent $n = 2$ and provides a natural baseline from which exponent deformations may be measured.

The Local Closure Defect, δ_K . To quantify the effect of exponent deformation on the Euclidean Target, the closure-defect functional introduced in Section 2 is evaluated specifically at c_2 .

Definition. The Local Closure Defect is defined by

$$\delta_K(a, b) = d_P(c_2, n). \quad (8)$$

Substituting the definition of c_2 into the closure-defect functional yields

$$\delta_K(a, b) = \left| 1 - \frac{a^{2+K_P} + b^{2+K_P}}{c_2^{2+K_P}} \right|. \quad (9)$$

Using Equation (7), this may be written equivalently as

$$\delta_K(a, b) = \left| 1 - \frac{a^{2+K_P} + b^{2+K_P}}{(a^2 + b^2)^{(2+K_P)/2}} \right|. \quad (10)$$

The quantity δ_K isolates the localized structural strain experienced by the Euclidean Target as the exponent departs from its classical baseline value.

Several observations follow immediately:

1. $\delta_K \geq 0$ for all admissible parameter values.
2. $\delta_K = 0$ when $K_P = 0$, corresponding to the classical Euclidean closure relation.
3. Small values of $|K_P|$ produce localized deviations from perfect closure that may be studied through asymptotic analysis.

Unlike global obstruction measures, δ_K is explicitly local. It is therefore insensitive to the asymptotic approximation effects that motivate the present formulation. The remainder of the

paper investigates the behavior of δ_K in a neighborhood of the Euclidean baseline and establishes a first-order asymptotic expansion governing its growth.

4 The First-Order Closure–Obstruction Asymptotic Expansion Theorem

Theorem. Let $(a, b) \in (\mathbb{R}^+)^2$ be a fixed pair of positive base coordinates. As the Power Deformation Parameter

$$K_P = n - 2 \tag{11}$$

varies smoothly in a local neighborhood of the Euclidean baseline ($K_P \rightarrow 0$), the Local Closure Defect δ_K evaluated at the Euclidean Target c_2 admits a local asymptotic expansion whose leading-order coefficient is proportional to the Shannon entropy $H(W)$ of the normalized base weights measured in nats.

Specifically,

$$\delta_K = \frac{H(W)}{2}|K_P| + O(K_P^2), \tag{12}$$

where

$$w_a = \frac{a^2}{a^2 + b^2} \tag{13}$$

and

$$w_b = \frac{b^2}{a^2 + b^2} \tag{14}$$

are the normalized base weights. The corresponding Shannon entropy is

$$H(W) = -(w_a \ln w_a + w_b \ln w_b). \tag{15}$$

The entropy functional employed throughout this paper is the classical Shannon entropy, originally introduced as a quantitative measure of information and uncertainty (Shannon, 1948). A comprehensive mathematical treatment of entropy and related information-theoretic quantities may be found in Cover and Thomas (2006). Since $a, b > 0$, both w_a and w_b lie strictly between 0 and 1, implying

$$H(W) > 0 \tag{16}$$

for all admissible base coordinates.

Proof. We begin by expressing the fraction appearing in the Local Closure Defect using the classical exponential identity

$$x^y = \exp(y \ln x), \tag{17}$$

which is standard in real analysis and asymptotic expansion theory (Rudin, 1976; Ahlfors, 1979).

For the a -component,

$$\begin{aligned}\frac{a^{2+K_P}}{(a^2 + b^2)^{(2+K_P)/2}} &= \frac{a^2}{a^2 + b^2} \left(\frac{a}{\sqrt{a^2 + b^2}} \right)^{K_P} \\ &= w_a \exp \left[\frac{K_P}{2} \ln w_a \right].\end{aligned}\tag{18}$$

Applying the same transformation to the b -component yields

$$\delta_K = \left| 1 - \left[w_a \exp \left(\frac{K_P}{2} \ln w_a \right) + w_b \exp \left(\frac{K_P}{2} \ln w_b \right) \right] \right|.\tag{19}$$

We now expand the exponential functions about the Euclidean baseline $K_P = 0$ using the Maclaurin expansion

$$\exp(x) = 1 + x + O(x^2)\tag{20}$$

(Whittaker & Watson, 1927). Thus,

$$\exp \left[\frac{K_P}{2} \ln w_a \right] = 1 + \frac{K_P}{2} \ln w_a + O(K_P^2)\tag{21}$$

and

$$\exp \left[\frac{K_P}{2} \ln w_b \right] = 1 + \frac{K_P}{2} \ln w_b + O(K_P^2).\tag{22}$$

Substituting these expansions into Equation (19) gives

$$\delta_K = \left| 1 - \left[w_a \left(1 + \frac{K_P}{2} \ln w_a \right) + w_b \left(1 + \frac{K_P}{2} \ln w_b \right) \right] \right| + O(K_P^2).\tag{23}$$

Using the normalization identity

$$w_a + w_b = 1,\tag{24}$$

we obtain

$$\delta_K = \left| 1 - \left[1 + \frac{K_P}{2} (w_a \ln w_a + w_b \ln w_b) \right] \right| + O(K_P^2).\tag{25}$$

The constant terms cancel, yielding

$$\delta_K = \left| -\frac{K_P}{2} (w_a \ln w_a + w_b \ln w_b) \right| + O(K_P^2).\tag{26}$$

By the definition of Shannon entropy,

$$H(W) = -(w_a \ln w_a + w_b \ln w_b).\tag{27}$$

Substituting Equation (27) into Equation (26) gives

$$\delta_K = \frac{H(W)}{2} |K_P| + O(K_P^2).\tag{28}$$

Therefore, the Local Closure Defect admits a first-order asymptotic expansion whose leading-order coefficient is proportional to the Shannon entropy of the normalized base weights.

□

Remark 1. Theorem 1 is a local asymptotic result valid in a neighborhood of the Euclidean baseline

$$K_P = 0. \quad (29)$$

It should not be interpreted as a global obstruction theorem, nor as a statement concerning the existence or non-existence of integer solutions for arbitrary exponents. The theorem characterizes the local behavior of the Local Closure Defect in a neighborhood of the Euclidean baseline and establishes only the first-order asymptotic behavior of the deformation.

Table 1 presents numerical values of the exact Local Closure Defect δ_K , the first-order entropy approximation $\frac{H(W)}{2}|K_P|$, and the corresponding residual errors for the representative base coordinates $(a, b) = (3, 4)$.

Table 1: Numerical Behavior of the Local Closure Defect and First-Order Entropy Approximation for $a = 3$ and $b = 4$ ($H(W) = 0.65342$ nats).

K_P	n	Exact δ_K	$(H(W)/2) K_P $	Residual Error
-0.20	1.80	0.06793	0.06534	0.00259
-0.10	1.90	0.03331	0.03267	0.00064
-0.05	1.95	0.01649	0.01634	0.00016
0.00	2.00	0.00000	0.00000	0.00000
0.05	2.05	0.01618	0.01634	0.00016
0.10	2.10	0.03205	0.03267	0.00062
0.20	2.20	0.06290	0.06534	0.00244

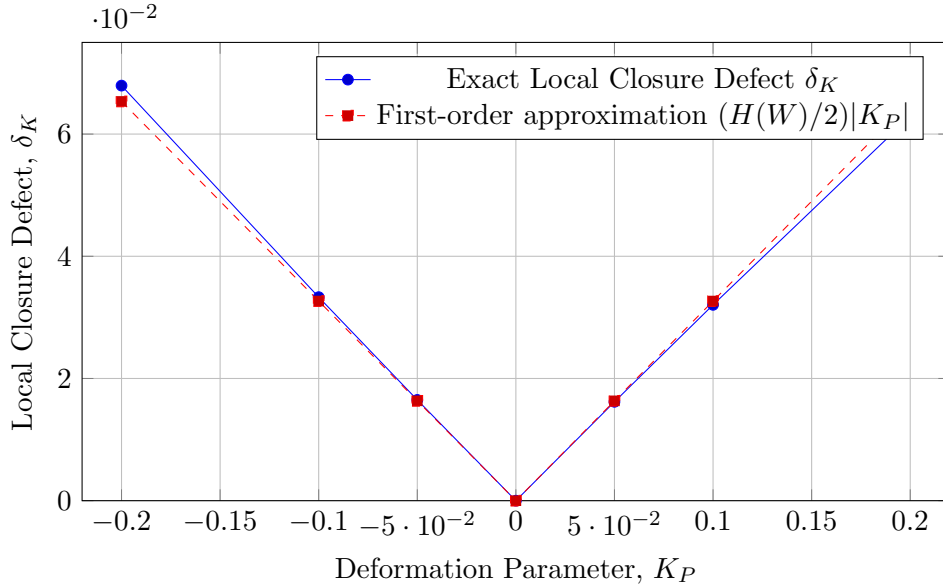


Figure 1: Local Asymptotic Behavior of the Closure Defect for $a = 3$ and $b = 4$.

Figure 1 compares the exact Local Closure Defect δ_K and the first-order asymptotic approximation $(H(W)/2)|K_P|$ for the representative base coordinates $a = 3$ and $b = 4$. The Euclidean baseline occurs at $K_P = 0$ ($n = 2$), where perfect closure is recovered. The close agreement be-

tween the exact defect and the asymptotic approximation near the baseline confirms the validity of the first-order expansion, while deviations for larger $|K_P|$ values arise from higher-order terms of order $O(K_P^2)$.

5 Formal Scope and Analytical Boundaries

To preserve mathematical clarity and avoid common over-interpretation pitfalls associated with new deformation frameworks, APG explicitly establishes its formal operational scope and analytical limitations.

5.1 Local versus Global Scope

Theorem 1 establishes a local continuous asymptotic expansion in a neighborhood of the Euclidean baseline

$$n = 2,$$

or equivalently

$$K_P = 0.$$

The theorem does not establish a positive global lower bound for any discrete obstruction functional and does not imply

$$\Omega_{\mathbb{Z}}(n) > 0 \tag{30}$$

throughout the full integer domain. Accordingly, APG should be interpreted as a local deformation framework rather than a global obstruction theory.

5.2 Motivation from Asymptotic Approximation

Global infimum constructions defined over infinite coordinate domains may approach zero through asymptotic approximation phenomena, including dense Diophantine near-misses (Bugeaud, 2004; Schmidt, 1980). This observation motivates the localized baseline formulation adopted in APG.

Importantly, this approximation behavior is not itself a theorem of APG. Rather, it serves as the analytical motivation for studying closure deformation locally around a fixed reference coordinate.

5.3 Distinction from Fermat’s Last Theorem

The APG framework neither provides nor implies an alternative proof of Fermat’s Last Theorem. The proof of Fermat’s Last Theorem relies on deep global structures involving elliptic curves, modular forms, Galois representations, and arithmetic geometry (Frey, 1986; Ribet, 1990; Wiles, 1995). Those methods operate at a fundamentally different level of mathematical structure than the local asymptotic analysis developed here.

The contribution of APG is more modest and more specific. The framework isolates a continuous deformation mechanism centered on the Euclidean baseline. As the exponent departs from

$$n = 2,$$

the Euclidean Target accumulates a measurable Local Closure Defect. Theorem 1 shows that the initial growth of this defect possesses a leading-order coefficient determined by the Shannon entropy of the normalized base weights.

Consequently, APG should be understood as a local analytical framework for studying exponent deformation and closure behavior rather than as a theory intended to resolve classical global Diophantine problems.

6 Relationship to Classical Asymptotic Analysis

The First-Order Closure–Obstruction Asymptotic Expansion Theorem is fundamentally an asymptotic expansion result. Similar local expansions occur throughout real analysis, perturbation theory, asymptotic analysis, and deformation theory (Rudin, 1976; Ahlfors, 1979). Consequently, the existence of a first-order expansion alone does not constitute the principal contribution of the present framework.

The novelty of Arithmetic Power Geometry (APG) does not lie in the existence of a Taylor-type expansion itself. Rather, the novelty lies in the organizational framework through which such expansions are interpreted and connected to localized closure behavior.

Specifically, APG introduces the following structural components:

1. The Power-Deformation Space

$$\mathcal{P}(a, b) = \{(c, n) \in \mathbb{R}^+ \times \mathbb{R}^+ : c > 0, n > 0\},$$

which treats exponentiation as a continuous deformation process rather than a fixed algebraic operation.

2. The Euclidean Target

$$c_2 = \sqrt{a^2 + b^2},$$

which serves as a distinguished baseline coordinate from which exponent deformations are measured.

3. The Local Closure Defect

$$\delta_K(a, b) = d_P(c_2, n),$$

which quantifies localized departures from perfect Euclidean closure.

4. The interpretation of Shannon entropy

$$H(W) = -(w_a \ln w_a + w_b \ln w_b)$$

as the leading-order coefficient governing local closure deformation.

Within this framework, the theorem establishes that

$$\delta_K = \frac{H(W)}{2}|K_P| + O(K_P^2)$$

in a neighborhood of the Euclidean baseline.

Accordingly, APG should be viewed as an axiomatic local deformation framework rather than as a replacement for arithmetic geometry, Diophantine geometry, or classical asymptotic analysis. The framework provides an alternative analytical perspective on exponent deformation and closure behavior while remaining fully compatible with established mathematical theories.

7 Future Research Program

To transition from an axiomatic local deformation framework to a broader mathematical theory, several directions merit future investigation.

7.1 Geometric Extensions of the Parameter Space

The present work employs a scalar closure-defect functional

$$d_P(c, n) = \frac{|c^n - a^n - b^n|}{c^n} \quad (31)$$

rather than a metric tensor.

Future work may investigate whether the Power-Deformation Space can be endowed with a non-degenerate metric structure inspired by information geometry (Amari, 2016) or differential geometry (Do Carmo, 2016). Only after such a structure is rigorously constructed would concepts such as curvature tensors, connections, or geodesics become mathematically meaningful within the framework.

7.2 Multi-Variable Deformation Models

The present analysis focuses on two-coordinate closure relations. A natural extension is the study of higher-dimensional systems such as

$$a^n + b^n + c^n = d^n \quad (32)$$

and related multi-variable closure structures.

In such settings, the behavior of the entropy coefficient

$$\frac{H(W)}{2} \quad (33)$$

may reveal new relationships between information-theoretic quantities and higher-dimensional closure phenomena.

7.3 Applications to Diophantine Parameter Spaces

Future work may also investigate localized deformation behavior in other Diophantine settings. Potential examples include:

- Pell-type equations,

- Catalan–Mihăilescu structures,
- generalized exponential Diophantine systems,
- rational approximation landscapes.

The objective would not be to replace existing arithmetic theories but to determine whether localized closure-defect analysis provides useful supplementary information regarding near-closure behavior and parameter sensitivity.

7.4 Information-Theoretic Interpretations

The appearance of Shannon entropy as the leading-order coefficient in Theorem 1 suggests a broader relationship between information-theoretic quantities and exponent deformation. The interpretation of entropy as an organizing principle for inference and structural description is consistent with the maximum-entropy viewpoint developed by Jaynes (1957). A systematic investigation of this connection may reveal whether other entropy measures, divergence functions, or information-geometric invariants play a role in governing localized closure behavior.

Related divergence-based quantities play a central role in the development of information geometry and statistical comparison (Kullback & Leibler, 1951).

8 Conclusion

Arithmetic Power Geometry (APG) provides a structured and internally consistent local deformation framework for investigating algebraic closure under continuous exponent variation. By treating the exponent as a deformation parameter and evaluating closure behavior relative to a fixed Euclidean baseline, the framework offers a localized analytical perspective on the behavior of power-type equations.

The central contribution of the paper is the First-Order Closure–Obstruction Asymptotic Expansion Theorem, which establishes that the Local Closure Defect admits a first-order asymptotic expansion in a neighborhood of the classical Euclidean baseline. The theorem demonstrates that the initial breakdown of classical Euclidean closure possesses a leading-order deformation coefficient determined by the Shannon entropy of the normalized base weights. This establishes a precise analytical connection between localized closure deformation and information-theoretic structure.

The framework is intentionally local in scope. It does not provide a global obstruction theorem, nor does it offer an alternative proof of Fermat’s Last Theorem. Rather, it introduces a mathematical language for studying exponent deformation, localized closure behavior, and entropy-controlled asymptotic strain.

Future work will determine whether these local structures can be extended to broader deformation spaces, higher-dimensional closure systems, or information-geometric formulations. Regardless of such developments, the present analysis establishes a rigorous first step toward understanding closure behavior through the lens of continuous exponent deformation.

“Numbers do not merely solve equations. They attempt to close worlds.”

References

- Ahlfors, L. V. (1979). *Complex Analysis* (3rd ed.). McGraw–Hill.
- Amari, S. I. (2016). *Information Geometry and Its Applications*. Springer.
- Bombieri, E., & Gubler, W. (2006). *Heights in Diophantine Geometry*. Cambridge University Press.
- Bugeaud, Y. (2004). *Approximation by Algebraic Numbers*. Cambridge University Press.
- Cover, T. M., & Thomas, J. A. (2006). *Elements of Information Theory* (2nd ed.). Wiley.
- Do Carmo, M. P. (2016). *Differential Geometry of Curves and Surfaces* (Revised and updated ed.). Dover Publications.
- Frey, G. (1986). Links between stable elliptic curves and certain Diophantine equations. *Annales Universitatis Saraviensis, Series Mathematicae*, 1(1), 1–40.
- Hartshorne, R. (1977). *Algebraic Geometry*. Springer–Verlag.
- Hindry, M., & Silverman, J. H. (2000). *Diophantine Geometry: An Introduction*. Springer–Verlag.
- Jaynes, E. T. (1957). Information theory and statistical mechanics. *Physical Review*, 106(4), 620–630.
- Kullback, S., & Leibler, R. A. (1951). On information and sufficiency. *Annals of Mathematical Statistics*, 22(1), 79–86.
- Lang, S. (2002). *Algebra* (3rd ed.). Springer–Verlag.
- Maclagan, D., & Sturmfels, B. (2015). *Introduction to Tropical Geometry*. American Mathematical Society.
- Ribet, K. A. (1990). On modular representations of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ arising from modular forms. *Inventiones Mathematicae*, 100(1), 431–476. <https://doi.org/10.1007/BF01231195>
- Rudin, W. (1976). *Principles of Mathematical Analysis* (3rd ed.). McGraw–Hill.
- Schmidt, W. M. (1980). *Diophantine Approximation*. Springer–Verlag.
- Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423.
- Silverman, J. H. (2007). *The Arithmetic of Elliptic Curves* (2nd ed.). Springer.
- Whittaker, E. T., & Watson, G. N. (1927). *A Course of Modern Analysis* (4th ed.). Cambridge University Press.
- Wiles, A. (1995). Modular elliptic curves and Fermat’s Last Theorem. *Annals of Mathematics*, 141(3), 443–551. <https://doi.org/10.2307/2118559>